



# Inteligencia Artificial y Aprendizaje Automático

Sesión 4: Redes Neuronales  
Convolucionales

Presentan:

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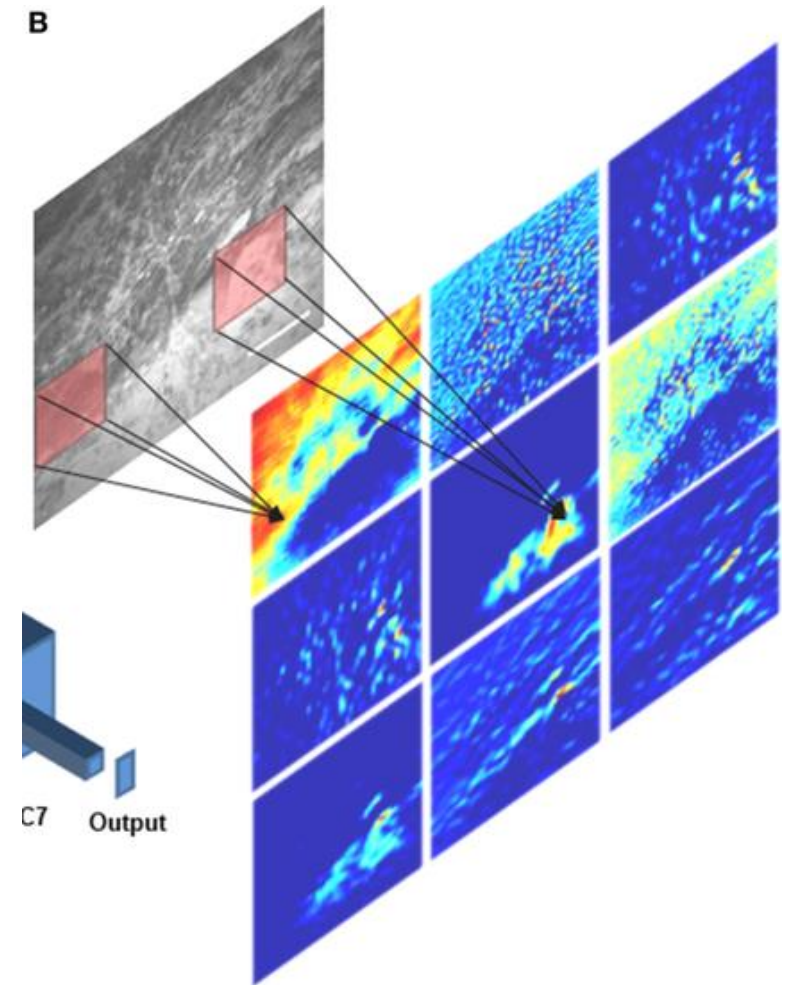


HPC  
DATA LAB

# Contenido

## Redes Neuronales convolucionales

- Definiciones
- Convolución
- Diseño de arquitecturas
- Ejercicios



# Las imágenes son números



|     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 157 | 153 | 174 | 168 | 150 | 152 | 129 | 151 | 172 | 161 | 155 | 156 |
| 155 | 182 | 163 | 74  | 75  | 62  | 33  | 17  | 110 | 210 | 180 | 154 |
| 180 | 180 | 50  | 14  | 34  | 6   | 10  | 33  | 48  | 106 | 159 | 181 |
| 206 | 109 | 5   | 124 | 131 | 111 | 120 | 204 | 166 | 15  | 56  | 180 |
| 194 | 68  | 137 | 251 | 237 | 239 | 239 | 228 | 227 | 87  | 71  | 201 |
| 172 | 105 | 207 | 233 | 233 | 214 | 220 | 239 | 228 | 98  | 74  | 206 |
| 188 | 88  | 179 | 209 | 185 | 215 | 211 | 158 | 139 | 75  | 20  | 169 |
| 189 | 97  | 165 | 84  | 10  | 168 | 134 | 11  | 31  | 62  | 22  | 148 |
| 199 | 168 | 191 | 193 | 158 | 227 | 178 | 143 | 182 | 106 | 36  | 190 |
| 206 | 174 | 155 | 252 | 236 | 231 | 149 | 178 | 228 | 43  | 95  | 234 |
| 190 | 216 | 116 | 149 | 236 | 187 | 86  | 150 | 79  | 38  | 218 | 241 |
| 190 | 224 | 147 | 108 | 227 | 210 | 127 | 102 | 36  | 101 | 255 | 224 |
| 190 | 214 | 173 | 66  | 103 | 143 | 96  | 50  | 2   | 109 | 249 | 215 |
| 187 | 196 | 235 | 75  | 1   | 81  | 47  | 0   | 6   | 217 | 255 | 211 |
| 183 | 202 | 237 | 145 | 0   | 0   | 12  | 108 | 200 | 138 | 243 | 236 |
| 195 | 206 | 123 | 207 | 177 | 121 | 123 | 200 | 175 | 13  | 96  | 218 |

What the computer sees

|     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 157 | 153 | 174 | 168 | 150 | 152 | 129 | 151 | 172 | 161 | 155 | 156 |
| 155 | 182 | 163 | 74  | 75  | 62  | 33  | 17  | 110 | 210 | 180 | 154 |
| 180 | 180 | 50  | 14  | 34  | 6   | 10  | 33  | 48  | 106 | 159 | 181 |
| 206 | 109 | 5   | 124 | 131 | 111 | 120 | 204 | 166 | 15  | 56  | 180 |
| 194 | 68  | 137 | 251 | 237 | 239 | 239 | 228 | 227 | 87  | 71  | 201 |
| 172 | 105 | 207 | 233 | 233 | 214 | 220 | 239 | 228 | 98  | 74  | 206 |
| 188 | 88  | 179 | 209 | 185 | 215 | 211 | 158 | 139 | 75  | 20  | 169 |
| 189 | 97  | 165 | 84  | 10  | 168 | 134 | 11  | 31  | 62  | 22  | 148 |
| 199 | 168 | 191 | 193 | 158 | 227 | 178 | 143 | 182 | 106 | 36  | 190 |
| 206 | 174 | 155 | 252 | 236 | 231 | 149 | 178 | 228 | 43  | 95  | 234 |
| 190 | 216 | 116 | 149 | 236 | 187 | 86  | 150 | 79  | 38  | 218 | 241 |
| 190 | 224 | 147 | 108 | 227 | 210 | 127 | 102 | 36  | 101 | 255 | 224 |
| 190 | 214 | 173 | 66  | 103 | 143 | 96  | 50  | 2   | 109 | 249 | 215 |
| 187 | 196 | 235 | 75  | 1   | 81  | 47  | 0   | 6   | 217 | 255 | 211 |
| 183 | 202 | 237 | 145 | 0   | 0   | 12  | 108 | 200 | 138 | 243 | 236 |
| 195 | 206 | 123 | 207 | 177 | 121 | 123 | 200 | 175 | 13  | 96  | 218 |





Image



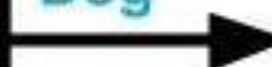
Neural Network

Cat



0.97

Dog



0.01

Other



0.02







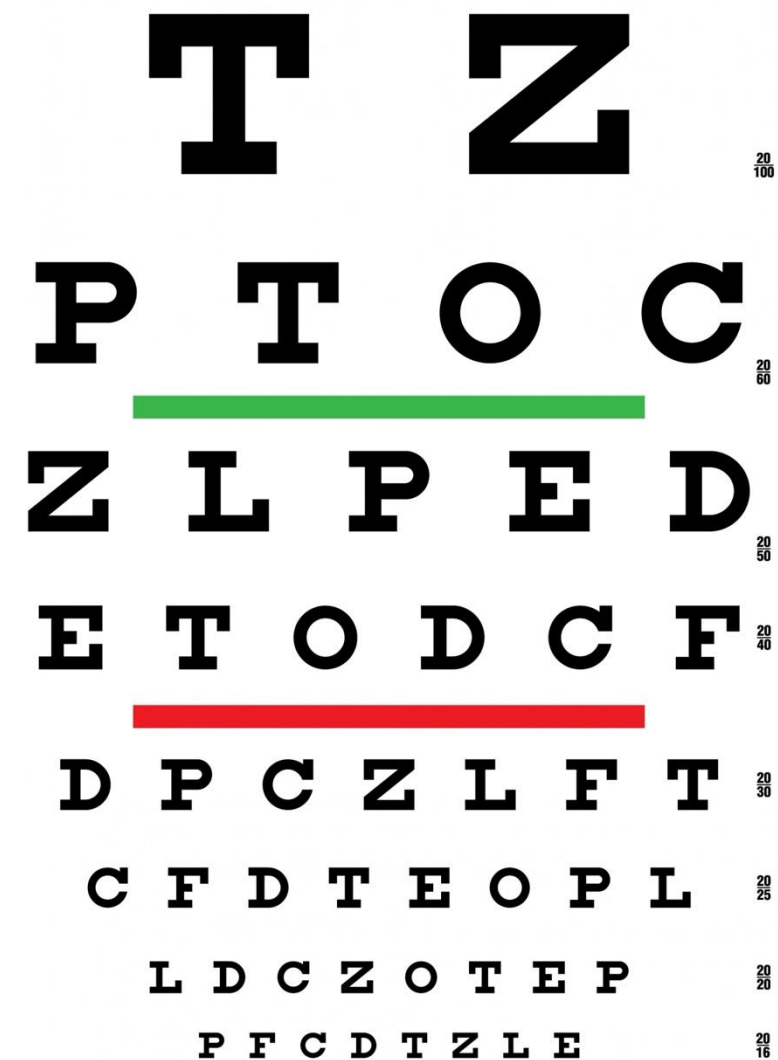




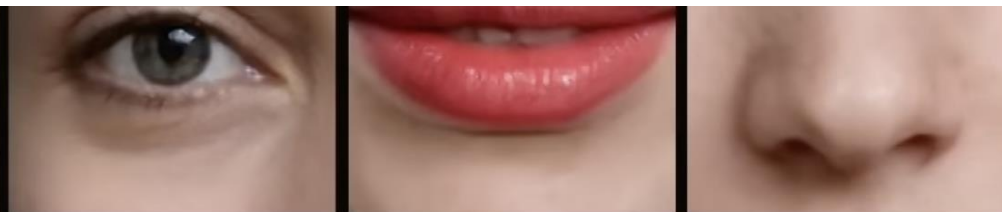
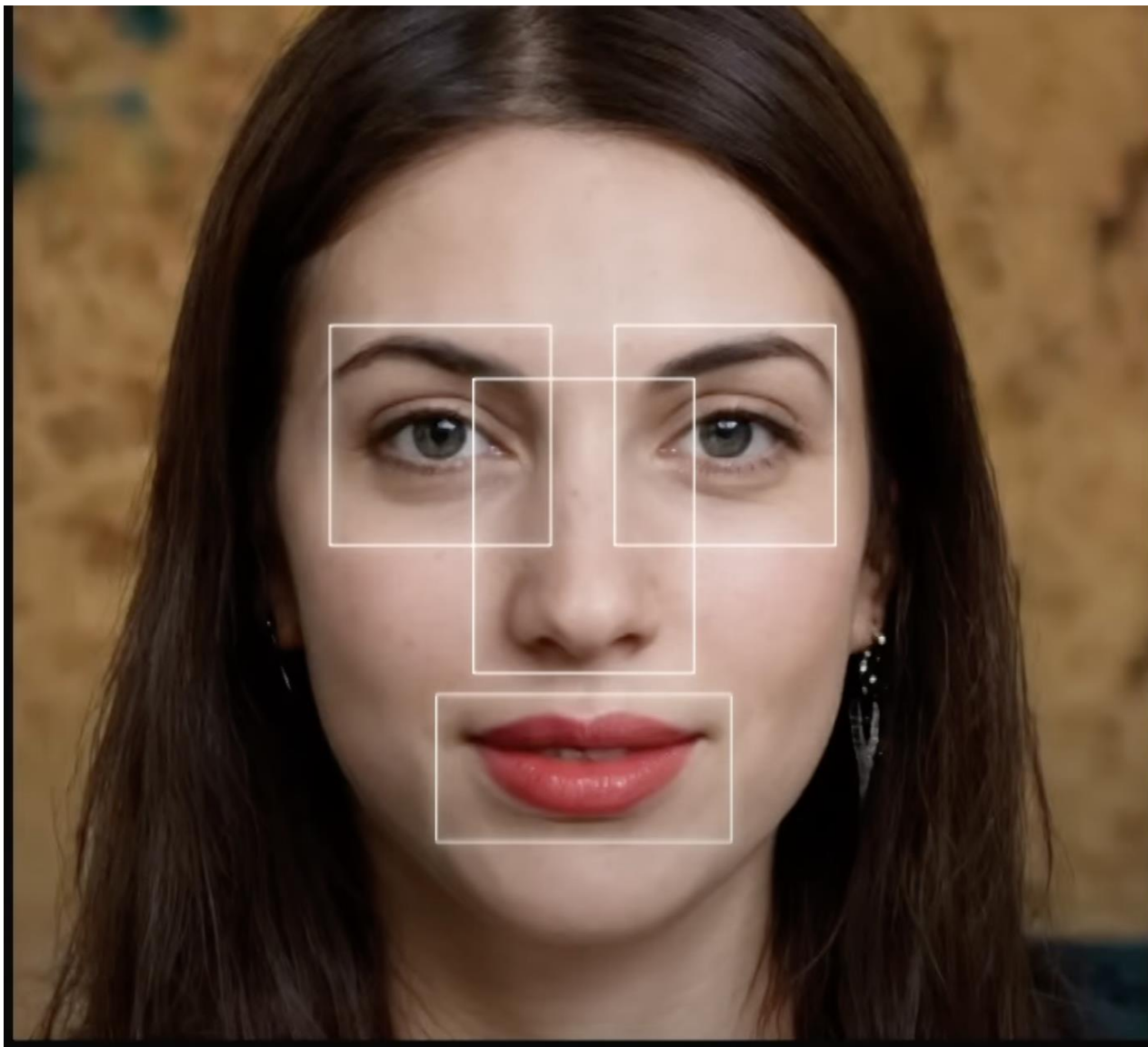
















# Redes Neuronales Convolucionales (Yann LeCun, 1989)



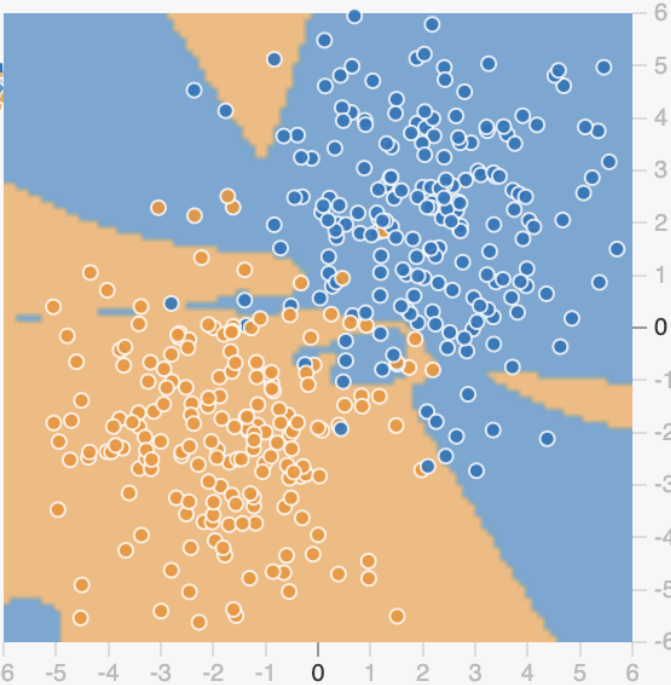
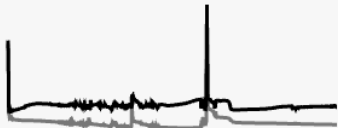
FEATURES

Which properties do you want to feed in?

+ - 5 HIDDEN LAYERS

OUTPUT

Test loss 0.150  
Training loss 0.048

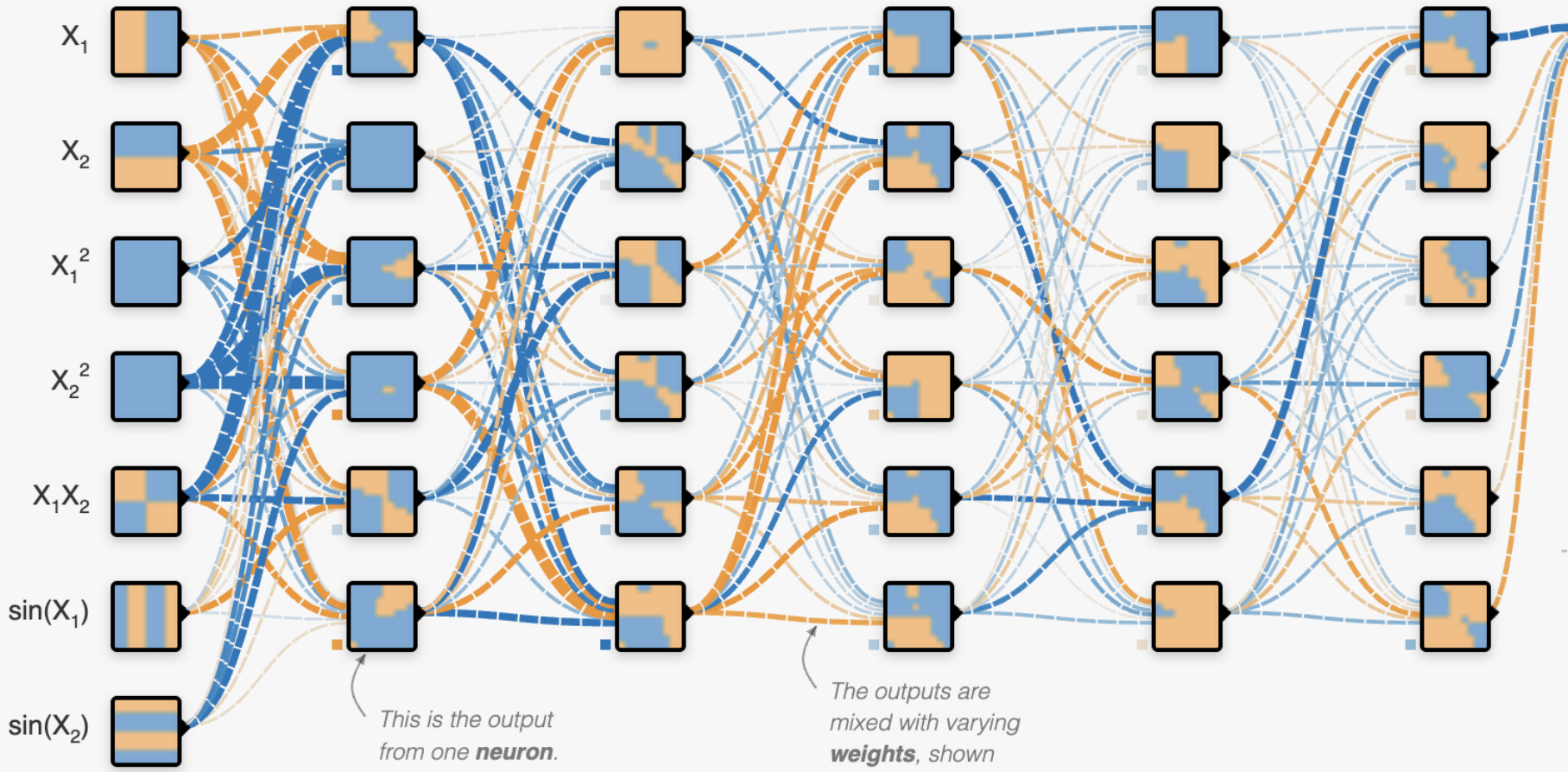


Colors shows data, neuron and weight values.

-1 0 1

☐ Show test data ☒ Discretize output

+ - 6 neurons  
+ - 6 neurons  
+ - 6 neurons  
+ - 6 neurons  
+ - 6 neurons



This is the output from one **neuron**.  
Hover to see it larger.

The outputs are mixed with varying **weights**, shown by the thickness of the lines.



28x28 pixels

`.flatten(img)` →



784x1 pixels

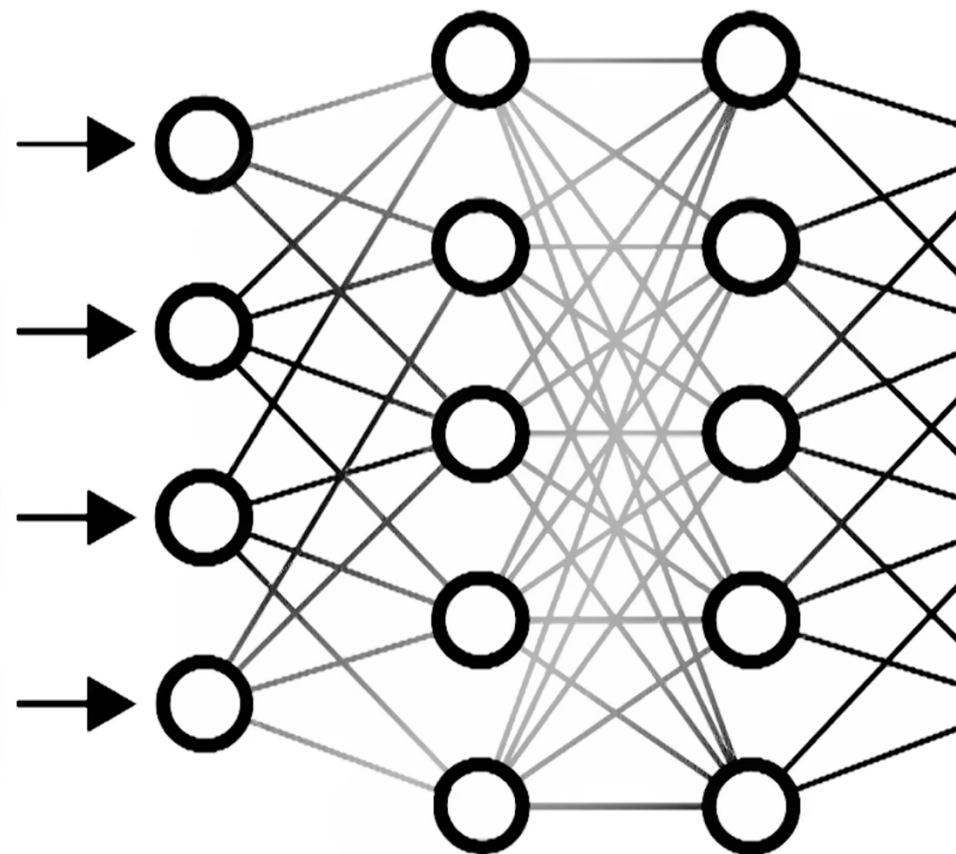
pixel 0

pixel 1

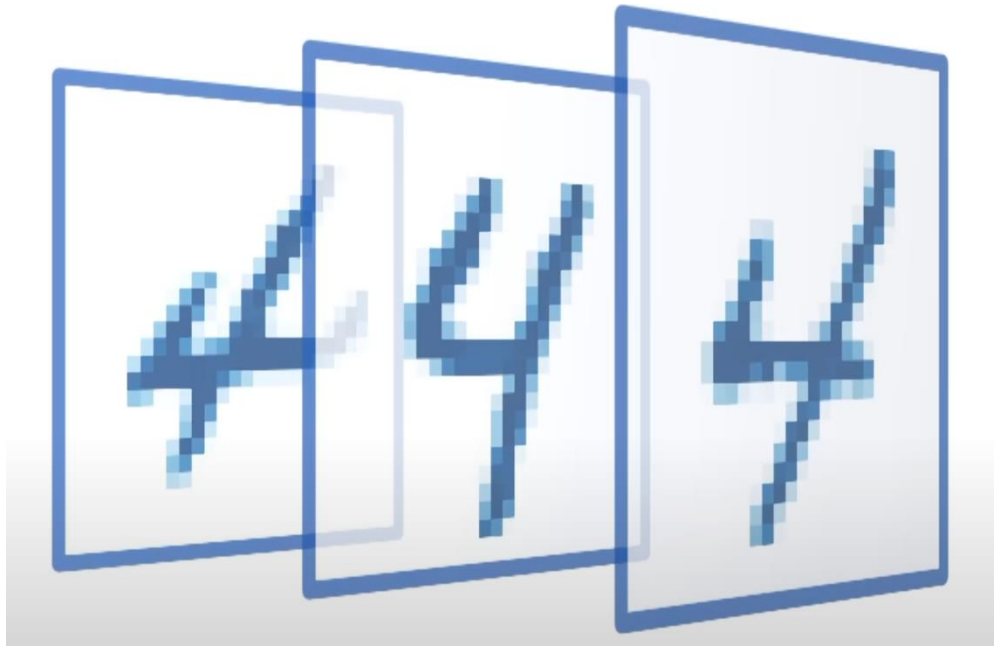
pixel 2

pixel 3

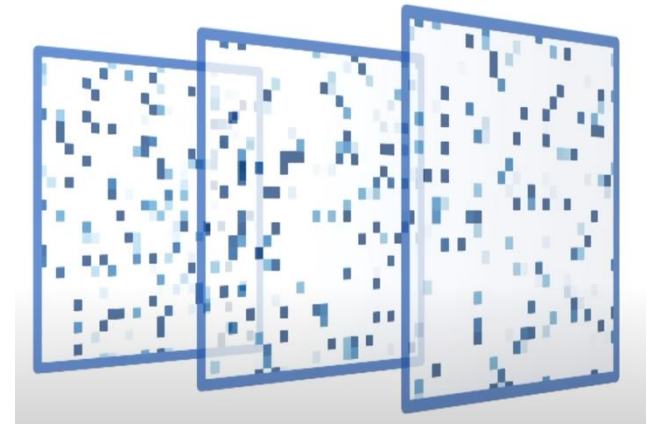
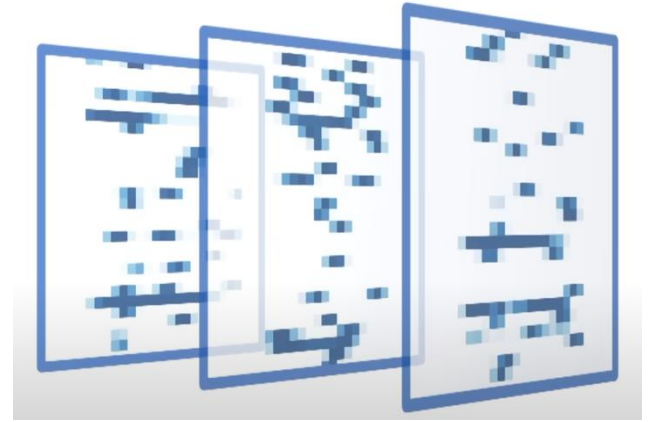
...



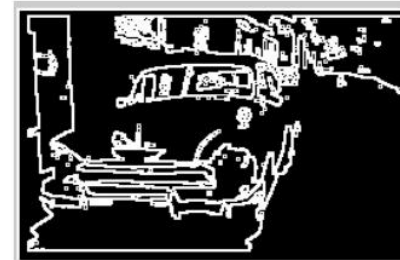
# Limitantes de las Redes Neuronales



=



# Filtros de convolución





Original



X-Pro II



Lomo-fi



Earlybird



Sutro



Lily



Toaster



Brannan



Inkwell



Walden



Hefe



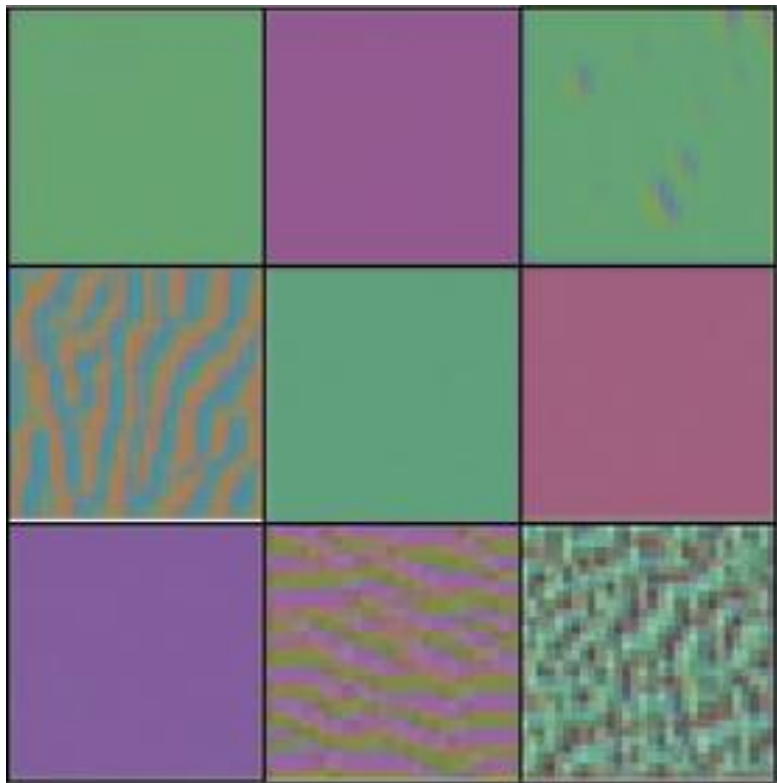
Apollo



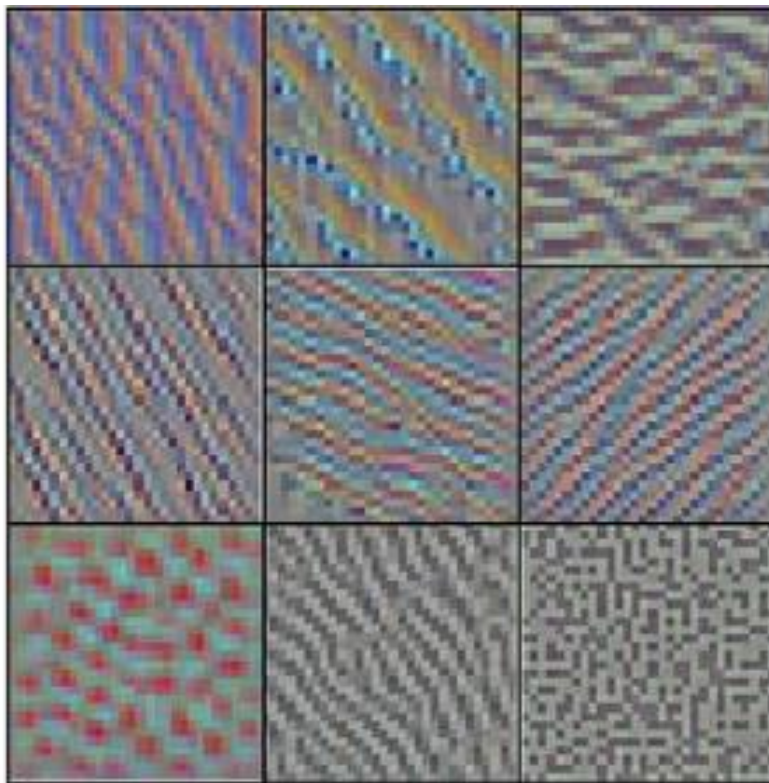
# Filtros de convolución

| <i>Original</i>                                                                    | <i>Gaussian Blur</i>                                                                | <i>Sharpen</i>                                                                       | <i>Edge Detection</i>                                                                |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$                | $\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$    | $\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$              | $\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$          |
|  |  |  |  |

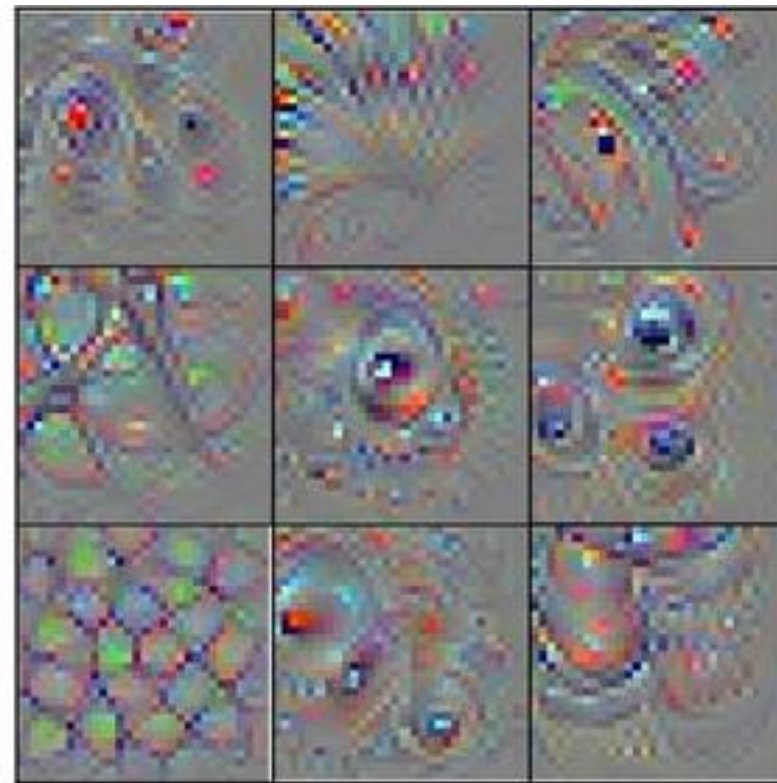
# Extracción de características



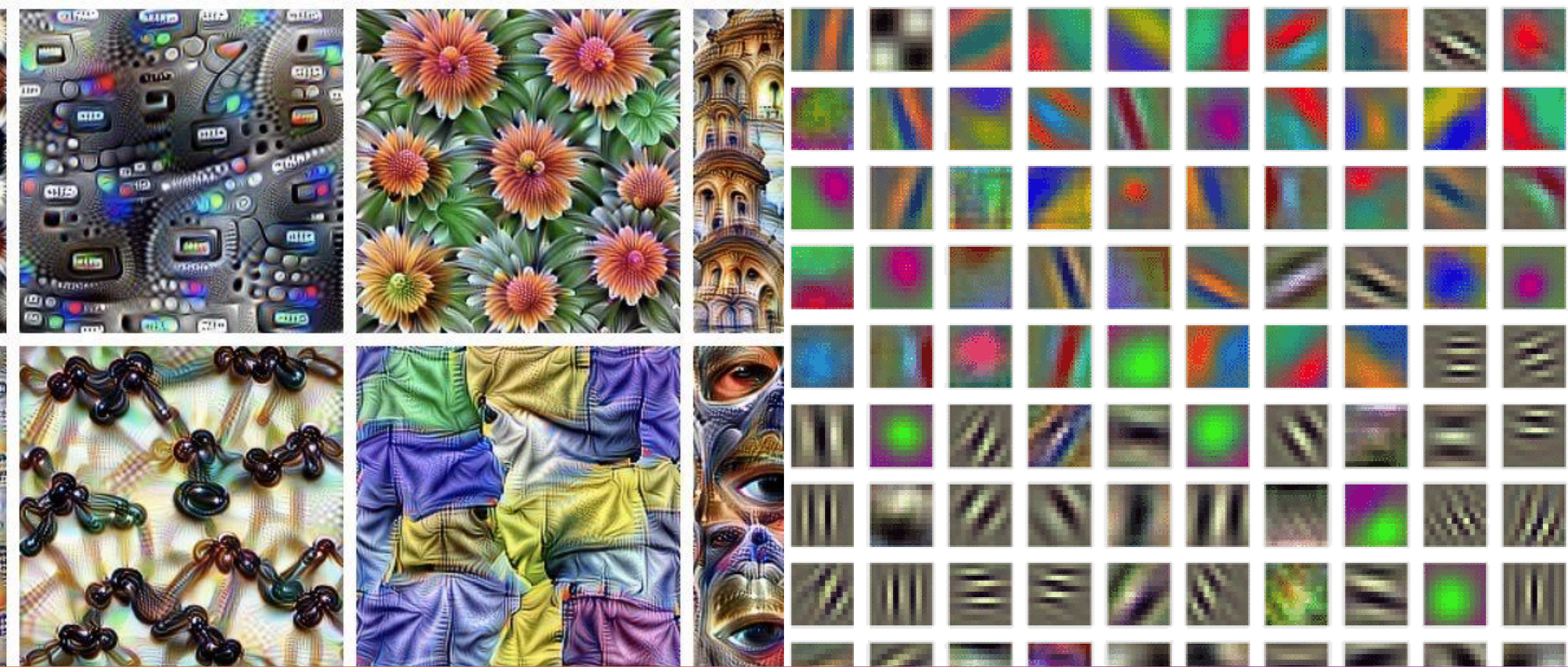
(a) Color



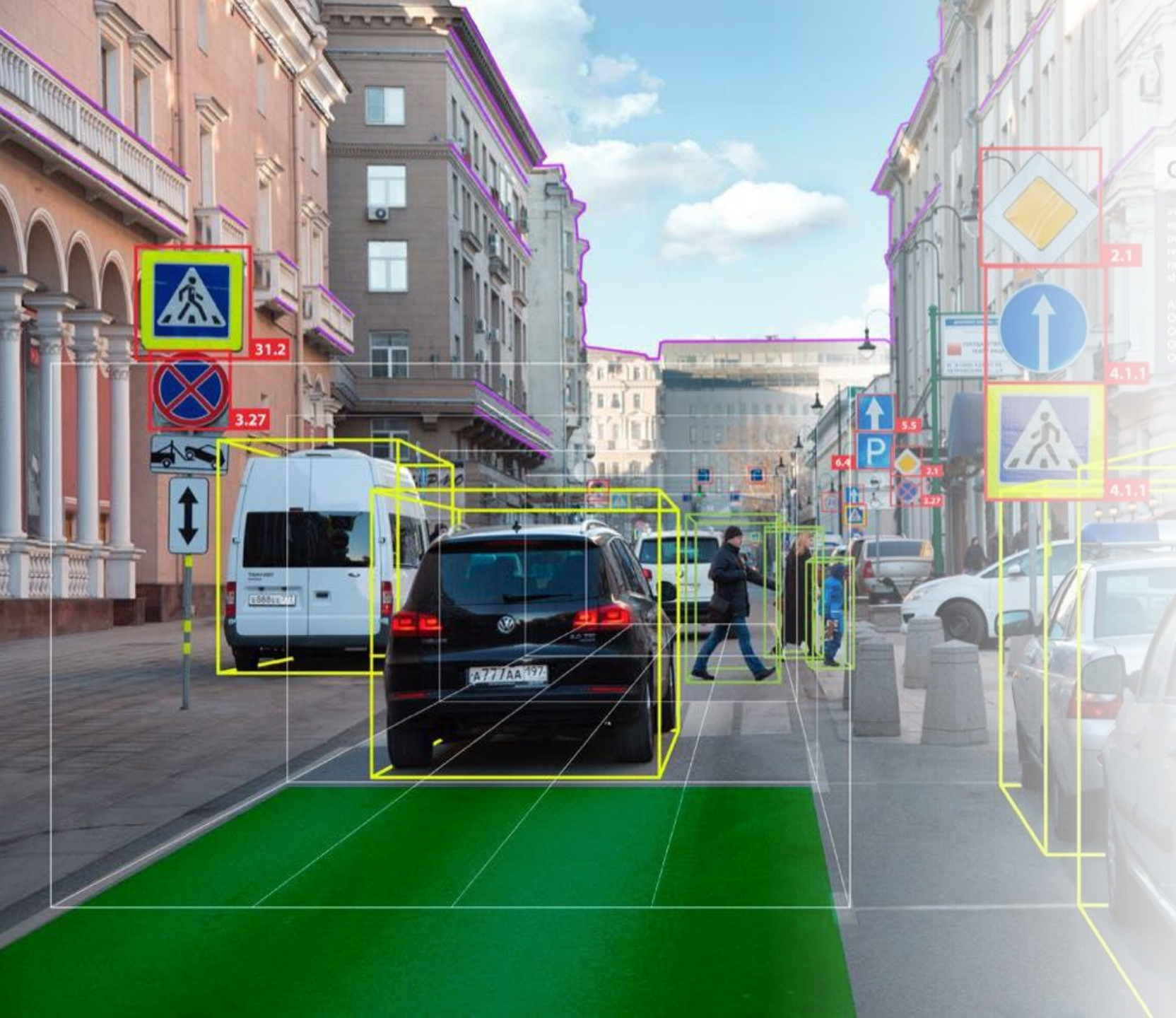
(b) Texture



(c) Structure



Filtros de convolución

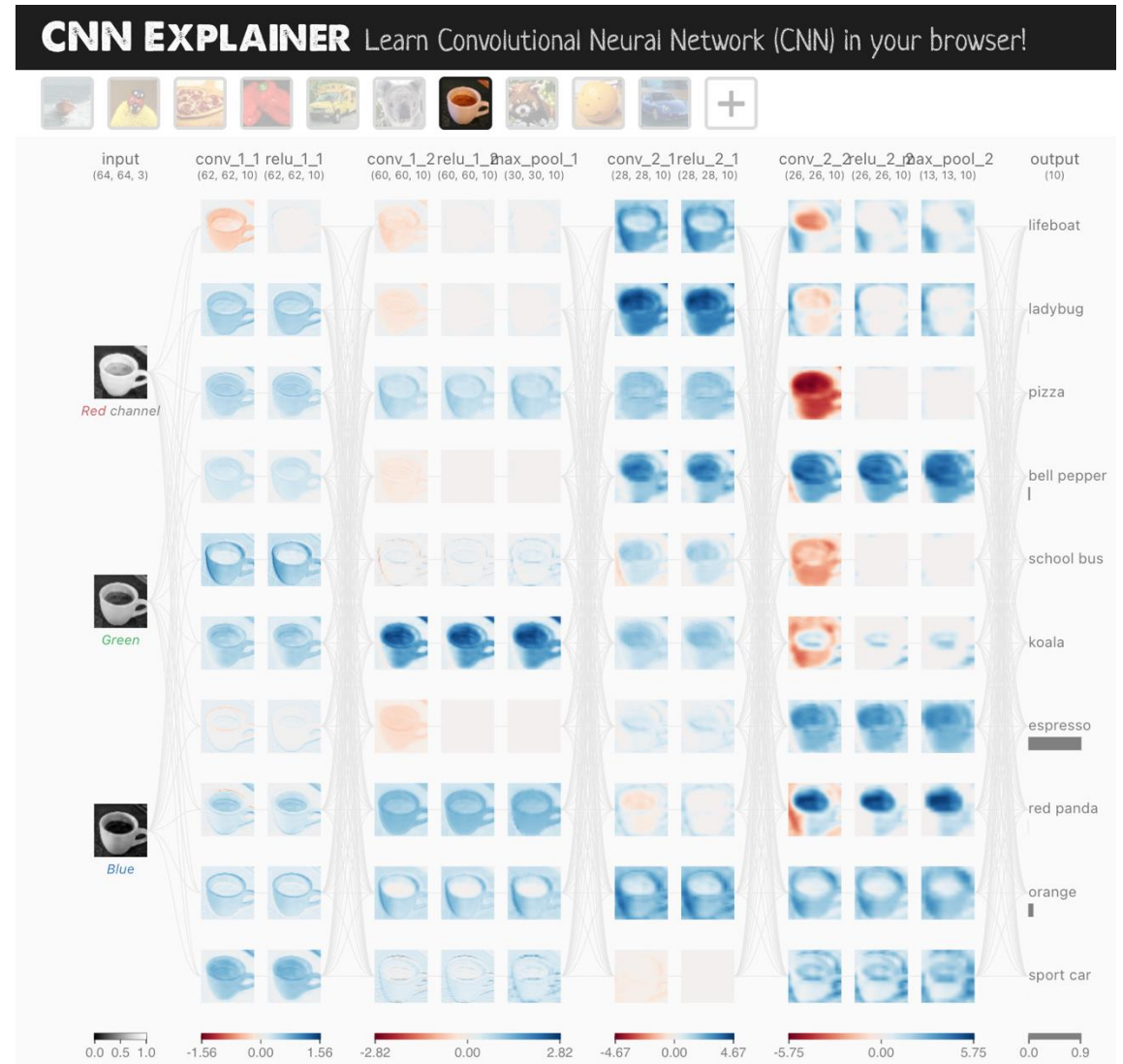


¿Dónde están los  
objetos, qué son los  
objetos, qué acciones  
están ocurriendo?

Predecir y anticipar  
eventos en el mundo

# CNN Explainer

- <https://poloclub.github.io/cnn-explainer/>



# Teacheble Machine

# Teachable Machine

<https://teachablemachine.withgoogle.com/>

### New Project

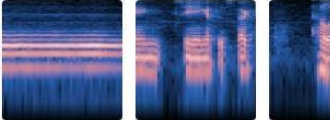
Open an existing project from Drive.

Open an existing project from a file.



#### Image Project

Teach based on images, from files or your webcam.



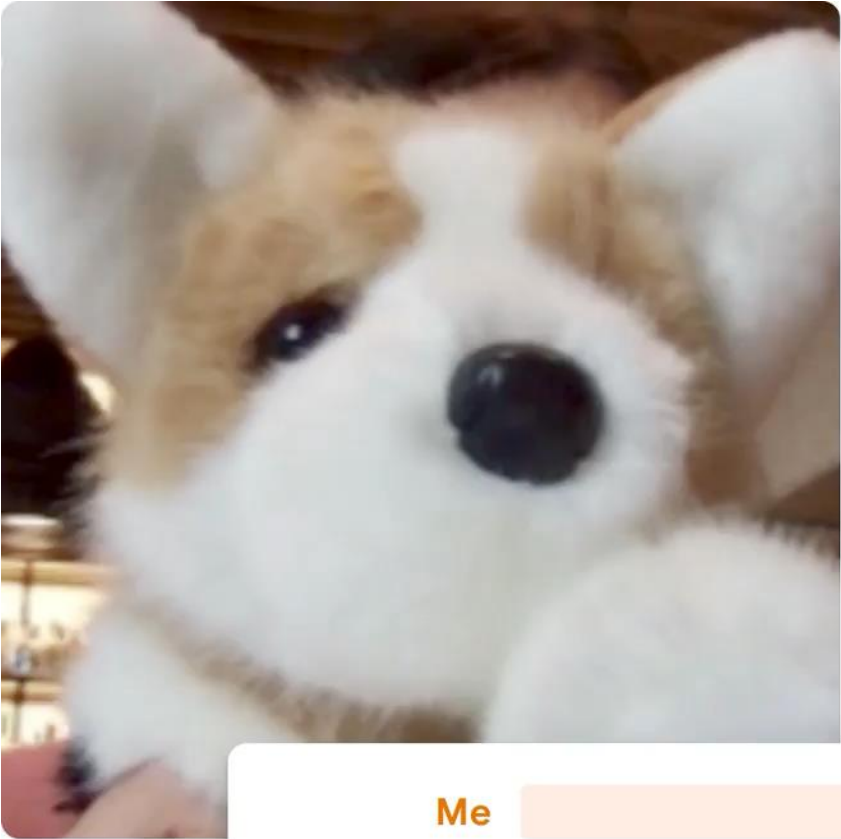
#### Audio Project

Teach based on one-second-long sounds, from files or your microphone.



#### Pose Project

Teach based on images, from files or your webcam.



Me

Me + Dog <3

20%